

Smart Mobility Solution Design: Intelligent Airport Security & Passenger Flow System

Name: Ankita Mukund Panpatil

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1. Problem Statement

As analyzed in Module 2, high-volume airport security checkpoints suffer from severe bottlenecking during peak departure banks. Manual document checks, inconsistent passenger preparation, and uncoordinated lobby arrivals lead to extended wait times, terminal overcrowding, and downstream flight delays.

2. Proposed Smart Mobility Solution

We propose the "**Intelligent Passenger Flow & Automated Security System (IPFASS)**". This solution integrates computer vision crowd-monitoring sensors, automated e-gates with facial recognition, 3D computed tomography (CT) scanners, and a real-time mobile app queue-management dashboard to streamline passenger throughput.

3. User and Operational Benefits

Benefit Category	Projected Improvement
Operational Efficiency	35% increase in passenger throughput per security lane and elimination of manual ID verification bottlenecks.
Passenger Experience	Reduced wait times, lowered anxiety via live queue time estimates on the airport mobile app, and faster security clearance.
Safety & Compliance	Enhanced threat detection accuracy through advanced CT scanning and automated biometric screening.

4. Implementation Requirements

- **Hardware Infrastructure:** Installation of 3D CT scanners, automated tray return loops, and biometric e-gates at security entry points.
- **Software & AI:** Computer vision crowd-sensing algorithms and cloud-based queue analytics integrated with airport operational databases.
- **Key Stakeholders:** Airport Operations Authority, TSA / Border Control agencies, Airline Ground Teams, and Technology Vendors.

5. Success Metrics

- **Average Queue Wait Time:** Target reduction of checkpoint wait times to under 15 minutes during peak hours.
- **Throughput Rate:** Achieving a sustained processing rate of 250+ passengers per lane per hour.
- **Passenger Satisfaction Score:** Elevating airport security experience ratings by at least 25%.

6. Solution Workflow & Visual Concept

IPFASS Operational Workflow

Step 1: Real-Time Flow Monitoring — IoT sensors detect lobby crowd density and push estimated wait times to the airport app.

Step 2: Biometric E-Gate Entry — Passengers clear preliminary identification via facial recognition without presenting paper boarding passes.

Step 3: Automated CT Scanning & Prep — Passengers place bags into automated smart trays; electronics remain inside bags during 3D scanning, eliminating manual bag searches.